HW2: advection of volume fraction

I heard you

- A large proportion of you are having trouble
  - Understanding the solver
  - Running the solver and conventions
  - Dealing with numeric
- All of which are results of not writing your own solver!

Reference Paper

- *Accurate conservative phase-field method for simulation of two-phase flows , Suhas S. Jain, 2022*
- You should be able to find all the equations, numerics, even the parameters needed for the spatial discretisation in this paper, along with cases to test on and compare against.
- The basic 1st order spatial discretisation is not in there but will be explained in this slide.
- The temporal discretisation will also be explained in this slide.

Structure of this lecture

- Scalar advection: 1st order
- CDI: 1st order
- CDI: 2nd order
- ACDI: 2nd order

Problem: advection of volume fraction

- Think of droplet (liquid in gas) or bubble (gas in liquid), that are being transported by the bulk flow.
- Examples being cavitation bubble

Simplification

- As you probably know, there exist complex interactions between the two phases: heat, mass, momentum transfer etc.
- Even momentum transfer (incompressible turbulence) alone can cause you plenty of trouble, like in HW1.
- Imagine coupling multiphase with turbulence, introduce compressibility, mass and energy transfer……

- But we are going to ignore all that for now.

Simplification

- In this homework, you prescribe a **fixed velocity field**, most likely analytical, u = u(x,y,t).
- You then take the volume fraction of one phase as a **passive scalar**: i.e. it does not affect anything else, u, p, nu etc.
- The only task here is, to advect this passive scalar accurately. Which you will see, is not as trivial as you might expect.

Scalar advection equation

- Consider volume fraction ϕ, the conservation equation for it is:
- $\frac{\partial \phi}{\partial t} + \nabla \cdot \left(\underline{\mathrm{u}} \phi \right) = 0$
- This is ‘physically correct’, as it simply states that ϕ follows the flow field without being generated or destroyed.
- However, this does not work in numerical simulation.

Non-regulated ϕ transport

- Consider semi-discretised form of transport equation in finite volume form:
- For a cell centred at i, j
- $V\frac{\partial \phi}{\partial t} + \Delta_y J_{x,i-1/2,\mathrm{j}} - \Delta_y J_{x,i+1/2,\mathrm{j}} + \Delta_x J_{y,\mathrm{i},j-1/2} - \Delta_x J_{y,\mathrm{i},j+1/2} = 0$ where V is volume, Δ_x and Δ_y are cell size in x and y, J_x and J_y are flux of ϕ in x and y
- This should be familiar, as it is pretty much the same for every quantity transported.

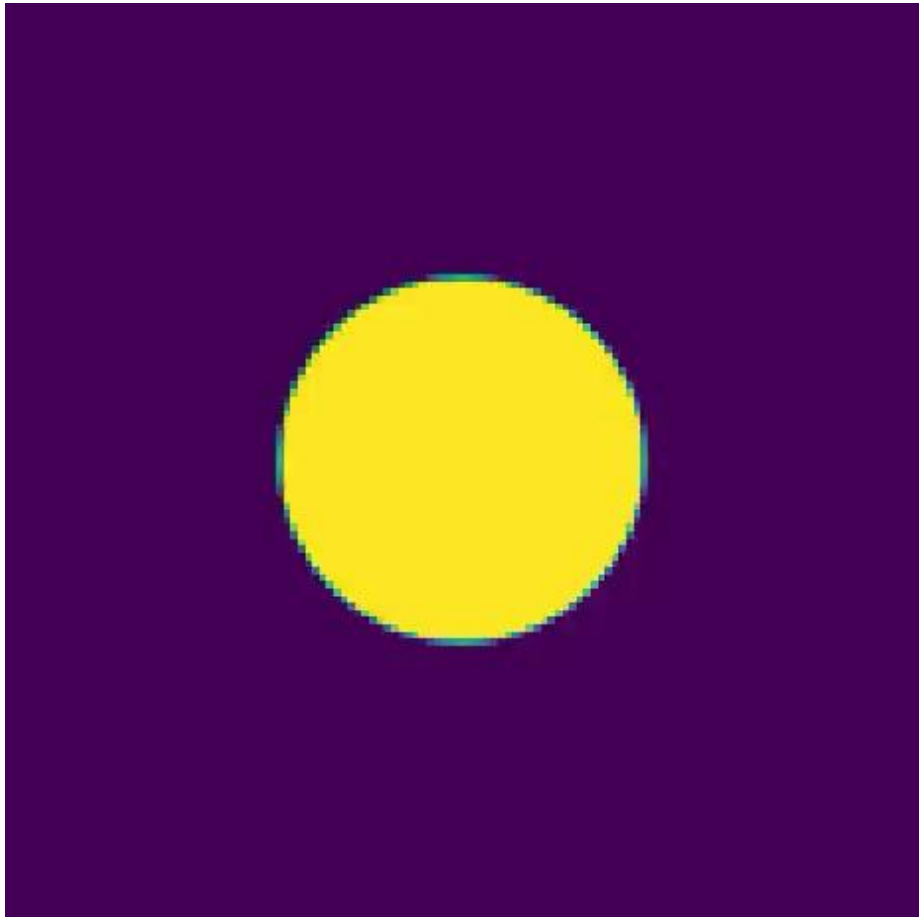
Non-regulated ϕ transport

- The problem arises when you try to formulate $J_{x,i-1/2}$ etc, i.e. the flux at cell faces.
- Using 1st order schemes introduce strong numerical diffusion
- Using higher (2nd) order schemes need some dissipation to be stable, as ϕ can be discontinuous at interfaces. (MUSCL, ENO etc)
- The problem is, dissipation is key to stability, but ruins sharp interface.

Non-regulated ϕ transport

- If you don't believe it, you can try it out: lets try a basic 1st order upwind scheme and see what happens.
- $J_{x,i+1/2,\mathrm{j}} = \begin{cases} u_{i+1/2,j}\phi_{i,j} & \text{if } u_{i+1/2,j} > 0 \\ u_{i+1/2,j}\phi_{i+1,j} & \text{if } u_{i+1/2,j} < 0 \end{cases}$
- Note that u is not upwinded as in this case u can be directly computed at cell face
- For temporal, just use 1st order explicit:
- $\frac{\partial \phi}{\partial t} = \frac{\phi^{n+1} - \phi^{n}}{\Delta t}$

Result of non-regulated ϕ transport



2nd order scheme helps, but only marginally

Interfacial regularisation

- Now lets consider the interfacial regularisation.
- This is an ‘artificial flux’ used to counter ‘artificial dissipation’
- There are two components to this: diffusion and sharpening. The two ‘balances out’ at an interface thickness that you prescribe.
- There are many formulations, for this HW we will test two of them.

Interfacial regularisation: Conservative Diffuse-Interface (CDI)

- Proposed by Chiu and Lin, 2011: *P.-H. Chiu, Y.-T. Lin, A conservative phase field method for solving incompressible two-phase flows, Journal of Computational Physics 230 (2011) 185–204*
- $\frac{\partial \phi}{\partial t} + \nabla \cdot \left(\underline{u}\phi\right) = \nabla \cdot \left\{\Gamma \left[\epsilon \nabla \phi - \phi(1-\phi)\frac{\nabla \phi}{|\nabla \phi|}\right]\right\}$
- Where Γ is a velocity scale and ϵ is the interface thickness scale

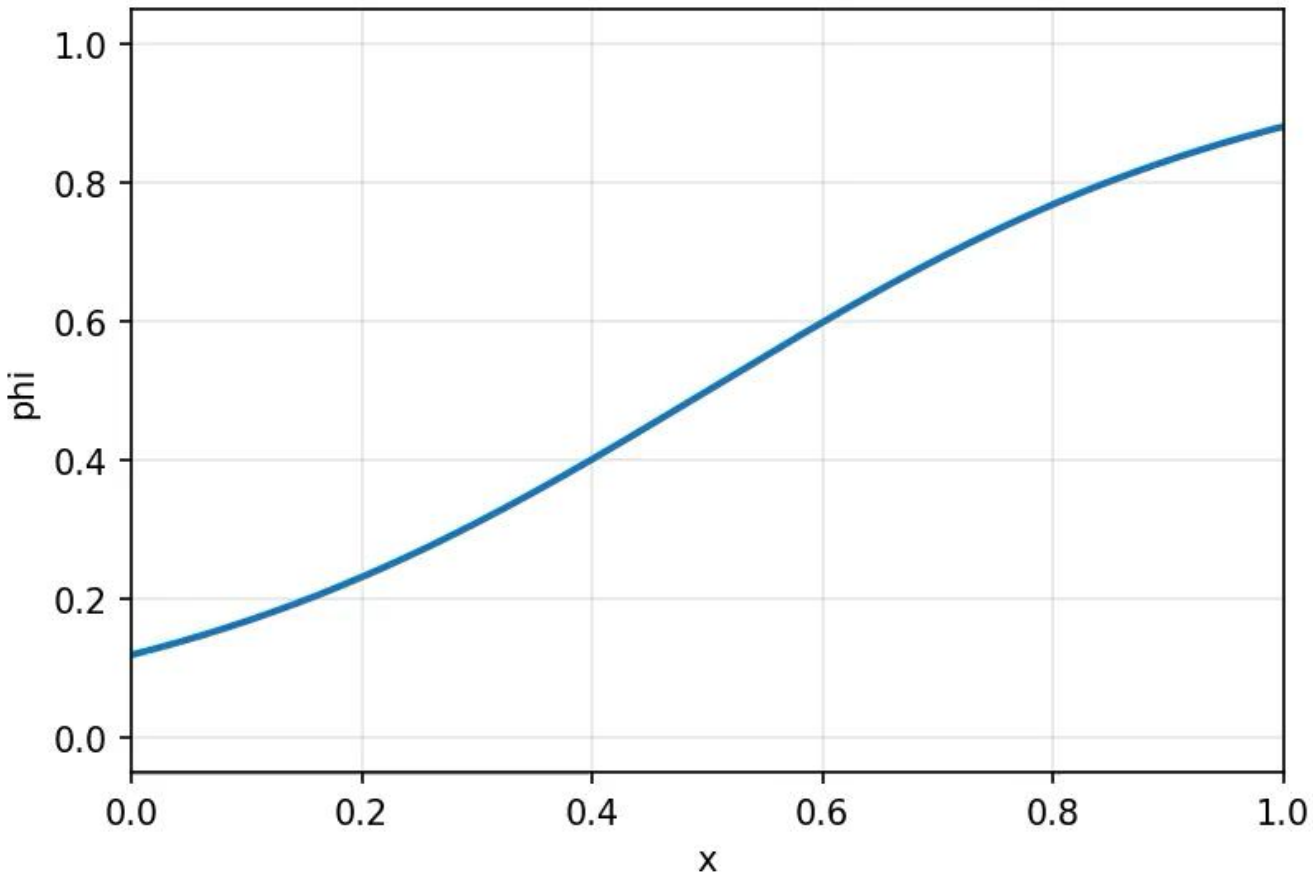
CDI: intuition

- Let's see what each term means:
- $\nabla \cdot \Gamma \epsilon \nabla \phi$: looks like $\nabla \cdot \nu \nabla \underline{\mathrm{u}}$?
- This is a diffusion term: instead of leaving stabilisation to numerical diffusion, we take it into our own hands.
- $-\nabla \cdot \Gamma \phi (1 - \phi) \frac{\nabla \phi}{|\nabla \phi|}$
- This is an interface sharpening term

CDI: sharpening term

- $-\nabla \cdot \Gamma\phi(1-\phi)\frac{\nabla\phi}{|\nabla\phi|}$
- When $\phi \longrightarrow 0$ or $\phi \longrightarrow 1$, this term vanishes
- $\phi(1-\phi)$ peaks at $\phi = 0.5$
- $\frac{\nabla\phi}{|\nabla\phi|}$ determines the direction normal the interface (parallel to $\nabla\phi$)
- Assuming $\frac{\partial\phi}{\partial \mathrm{x}} > 0$ in the outward direction, then this flux drives ϕ in the outwards direction, which sharpens the jump

CDI: sharpening term



CDI: Numerical Implementation

- Now we need to figure out how to compute this numerically
- The only difference is we now need to compute the regularisation flux

$$V\frac{\partial \phi}{\partial t} + (advective_terms) +$$
$$\Delta_y \mathrm{A}_{x,i-1/2,\mathrm{j}} - \Delta_y \mathrm{A}_{x,i+1/2,\mathrm{j}} + \Delta_x \mathrm{A}_{y,\mathrm{i},j-1/2} - \Delta_x \mathrm{A}_{y,\mathrm{i},j+1/2} = 0$$

- Where A are the regularisation fluxes

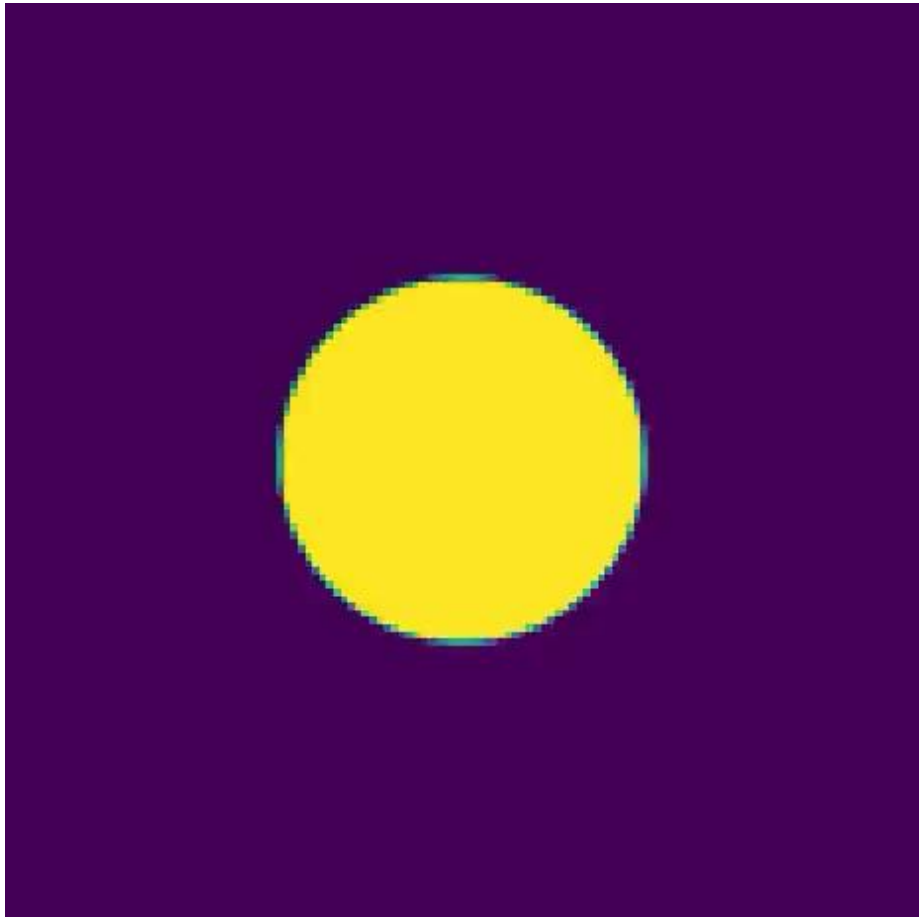
CDI: Numerical Implementation

- Gradient of ϕ at cell centre : $\left.\frac{\partial \phi}{\partial \mathrm{x}}\right|_{i,\mathrm{j}} = \frac{\phi_{i+1,\mathrm{j}} - \phi_{i-1,\mathrm{j}}}{2\Delta \mathrm{x}}$ (central 2nd order)
- Gradient of ϕ at cell face: $\left.\frac{\partial \phi}{\partial \mathrm{x}}\right|_{i+1/2,j} = \left(\left.\frac{\partial \phi}{\partial \mathrm{x}}\right|_{i,\mathrm{j}} + \left.\frac{\partial \phi}{\partial \mathrm{x}}\right|_{i,\mathrm{j}}\right)/2$ (average of the two)
- $\mathrm{A}_{x,i+1/2,\mathrm{j}} = \Gamma\left(\frac{\epsilon(\phi_{i+1,\mathrm{j}} - \phi_{i,\mathrm{j}})}{\Delta x} - \phi_{i+1/2,\mathrm{j}}(1 - \phi_{i+1/2,\mathrm{j}})\frac{\left.\frac{\partial \phi}{\partial \mathrm{x}}\right|_{i+1/2,j}}{|\nabla \phi|_{\mathrm{i}+1/2,\mathrm{j}}}\right)$
- Where $\phi_{i+1/2,\mathrm{j}} = \frac{\phi_{i+1,\mathrm{j}} - \phi_{i,\mathrm{j}}}{2}, |\nabla \phi|_{\mathrm{i}+1/2,\mathrm{j}} = \left.\sqrt{\frac{\partial \phi}{\partial \mathrm{x}}^2 + \frac{\partial \phi}{\partial \mathrm{y}}^2}\right|_{i,\mathrm{j}}$

CDI: stability criterion

- $\epsilon \geq 0.5\Delta\mathrm{x}$
- $\Gamma^* \geq \frac{1}{2\epsilon^* - 1}$ where $\Gamma^* = \frac{\Gamma}{|u|_{max}}$ and $\epsilon^* = \frac{\epsilon}{\Delta\mathrm{x}}$
- $\Delta t_{max} \sim \frac{\Delta\mathrm{x}}{|u|_{max}}\left(2 - \frac{1}{\epsilon^*}\right)$
- As we approach $\epsilon \sim 0.5\Delta\mathrm{x}$, Δt_{max} must be kept at a very small value. This is less than ideal.

CDI: does it work?



It is still very diffusive due to the first order scheme

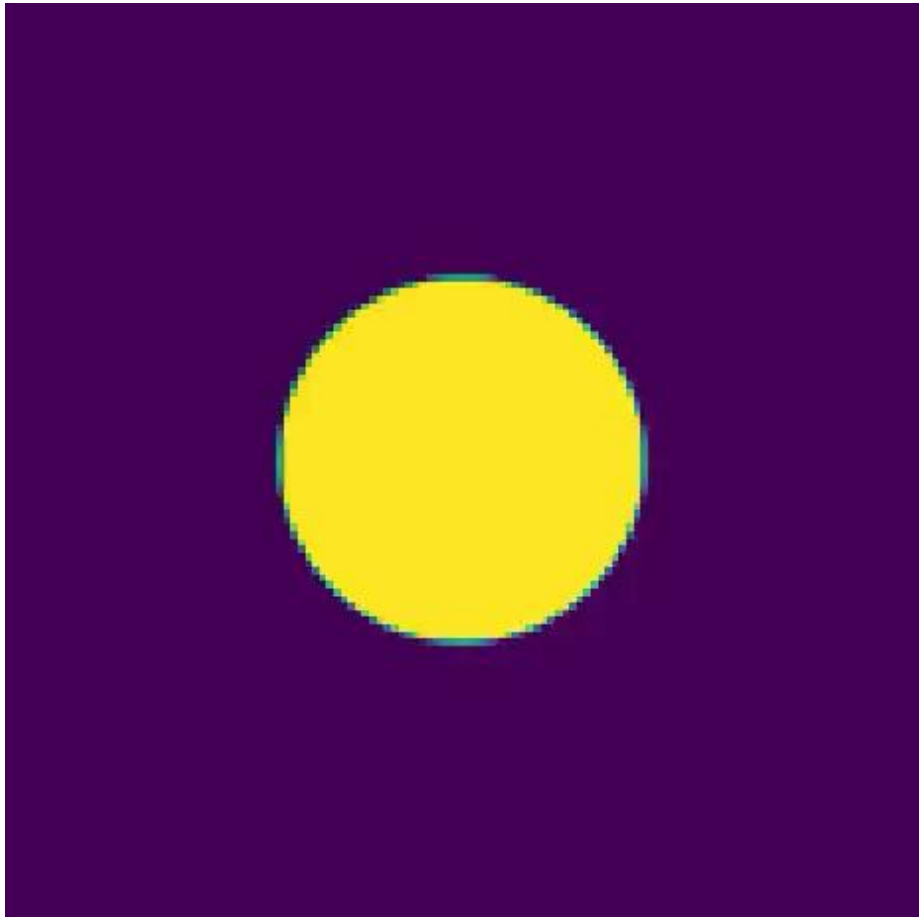
CDI: 2nd order implementation

- Instead of using upwind, we try central, as we already implemented artificial diffusion, we can use 2nd order scheme with stability.
- $J_{x,i+1/2,\mathrm{j}} = u_{i+1/2,j}(\phi_{i,j}+\phi_{i,j})/2$
- This result in much cleaner phase boundary.
- While we are here, might as well make the time stepping more accurate.

Runge-Kutta time stepping

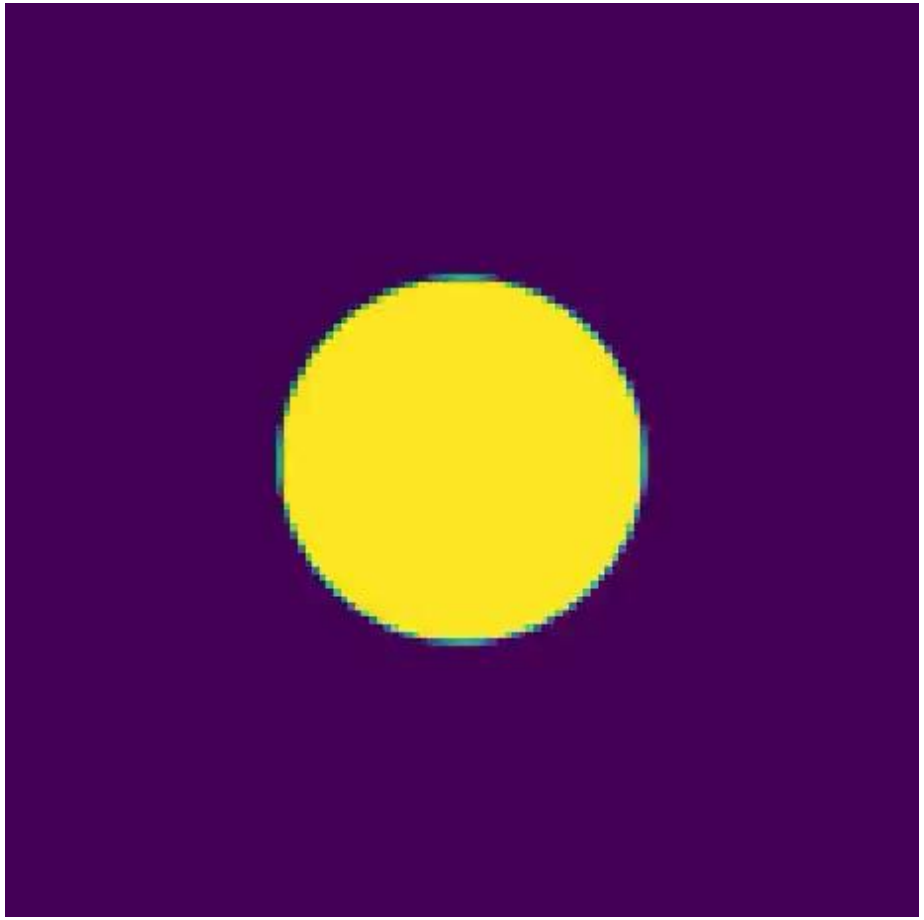
- A commonly used explicit time stepping method, easily integrated with various spatial discretisation.
- Consider $\frac{dy}{dt} = f(y, t), y(t = 0) = y_0$
- Time step size Δt
- $K_1 = f(t_n, y_n)$
- $K_2 = f(t_n + \Delta t/2, y_n + K_1 \Delta t/2)$
- $K_3 = f(t_n + \Delta t/2, y_n + K_2 \Delta t/2)$
- $K_4 = f(t_n + \Delta t, y_n + K_3 \Delta t)$
- $y_{n+1} = y_n + \Delta t/6(K_1 + 2K_2 + 2K_3 + K_4)$
- In our case, y is ϕ

CDI: 2nd order implementation



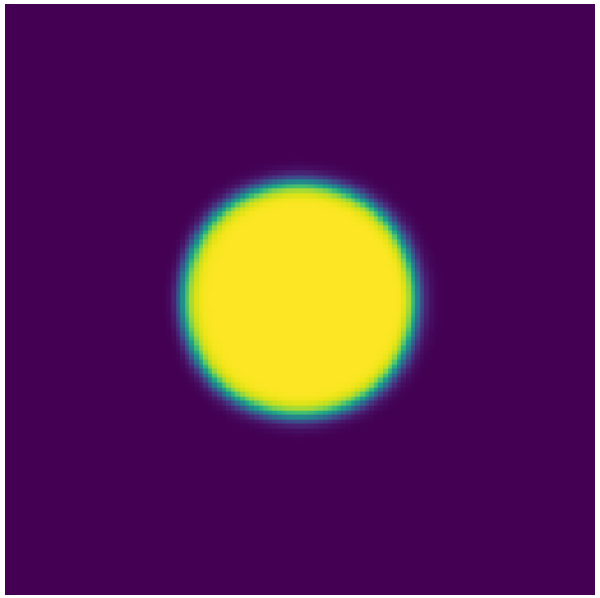
Kind of works!

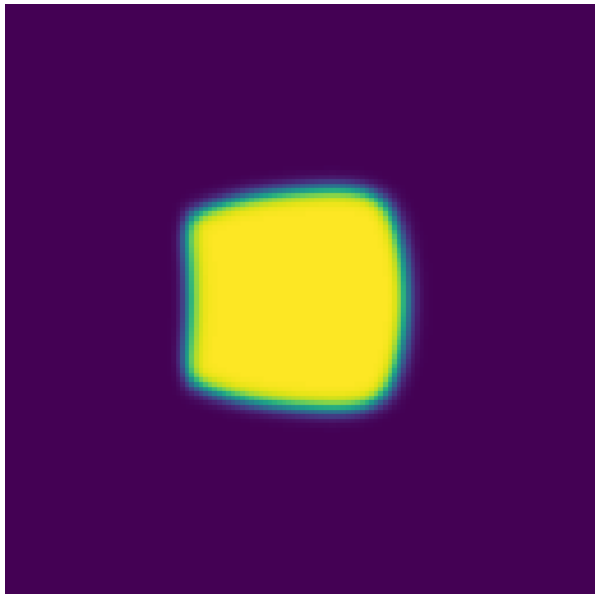
CDI: 2nd order implementation



But wait… what?

That isn’t quite a circle…





Fix to this (novelty of this paper) : Accurate CDI

- Intuition behind ACDI: instead of using ϕ to regulate interface, which introduce directionality and other numerical issue, we use ψ as an alternative.
- As the paper showed, ψ is linear with respect to the distance to interface, while ϕ is not, which introduces numerical issue.
- $\psi = \epsilon \ln(\frac{\phi}{1-\phi})$, but for numerical stability, a small value is added:
- $\psi = \epsilon \ln(\frac{\phi+\omega}{1-\phi+\omega}), \omega = 1e-100$

Transport equation in terms of ψ

- Original equation: $\frac{\partial \phi}{\partial t} + \nabla \cdot (\underline{u}\phi) = \nabla \cdot \left\{ \Gamma \left[\epsilon \nabla \phi - \phi(1-\phi) \frac{\nabla \phi}{|\nabla \phi|} \right] \right\}$
- New formulation replaces the interface sharpening term using ψ
- $\frac{\partial \phi}{\partial t} + \nabla \cdot (\underline{u}\phi) = \nabla \cdot \left\{ \Gamma \left[\epsilon \nabla \phi - \frac{1}{4} \left[1 - tanh^2 \left(\frac{\psi}{2\epsilon} \right) \right] \frac{\nabla \psi}{|\nabla \psi|} \right] \right\}$
- Note that the LHS and diffusion terms are kept as is!

Stability of ACDI

- As the paper discussed, the criterion for stability is:
- $\Gamma \geq \left|\underline{\mathrm{u}}\right|_{max}$ and $\epsilon \geq 0.5\Delta\mathrm{x}$
- Intuitively, Γ is 'how fast the diffused interface shifts towards equilibrium' ϵ is 'how thick the interface is'
- Setting them to their respective limit is usually preferred.
- CFL criterion is: $\Delta t_{max} \sim \frac{\Delta\mathrm{x}}{|u|_{max}\epsilon^*}$, this is less restrictive than CDI

ACDI: numerics

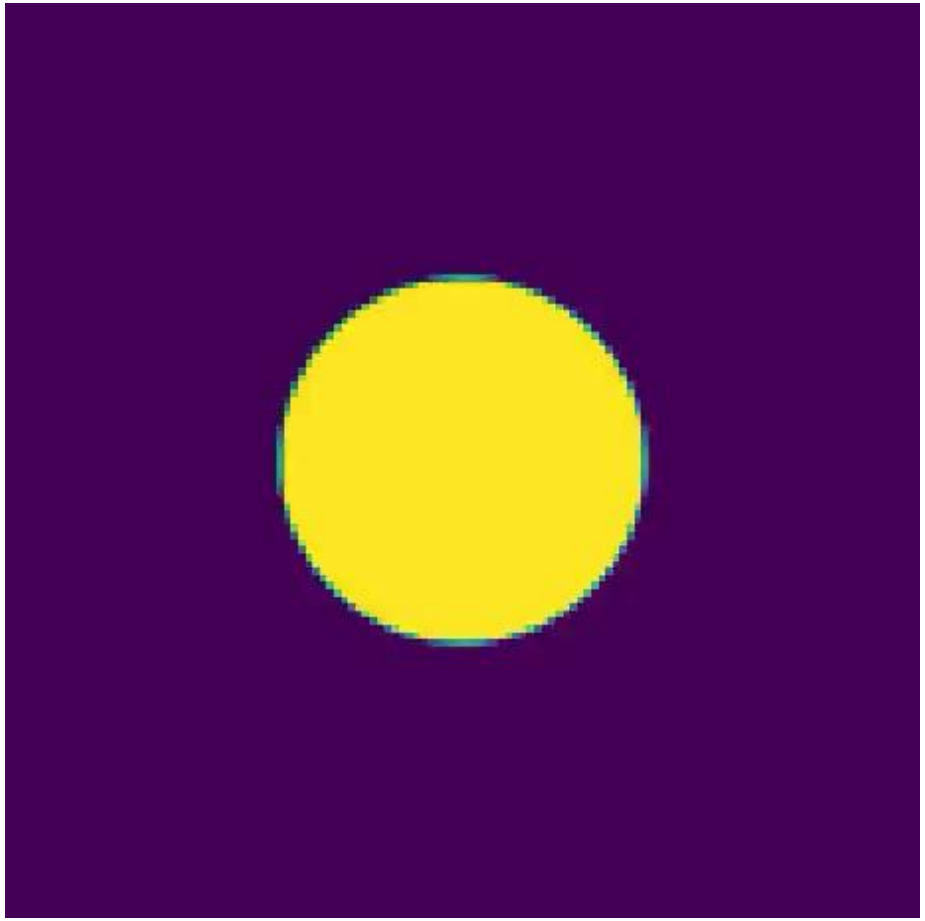
$$V\frac{\partial\phi}{\partial t} + (advective_terms) +$$
$$\Delta_y \mathrm{A}_{x,i-1/2,\mathrm{j}} - \Delta_y \mathrm{A}_{x,i+1/2,\mathrm{j}} + \Delta_x \mathrm{A}_{y,\mathrm{i},j-1/2} - \Delta_x \mathrm{A}_{y,\mathrm{i},j+1/2} = 0$$

- We only need to change the A terms
- $\mathrm{A}_{x,i+1/2,\mathrm{j}} = \left\{\Gamma\left\{\frac{\epsilon(\phi_{i+1,\mathrm{j}}-\phi_{i,\mathrm{j}})}{\Delta \mathrm{x}} - \frac{1}{4}\left[1 - tanh^2\left(\frac{\psi_{i+1/2}}{2\epsilon}\right)\right]\ \frac{\nabla\psi}{|\nabla\psi|}\bigg|_{i+1/2}\right\}\right\}$

- Gradient is calculated the same way as ϕ in CDI.

ACDI: test

Still kind of sqare-ish..... But at least symmetric.
More significant improvements appears when the droplet is deformed.

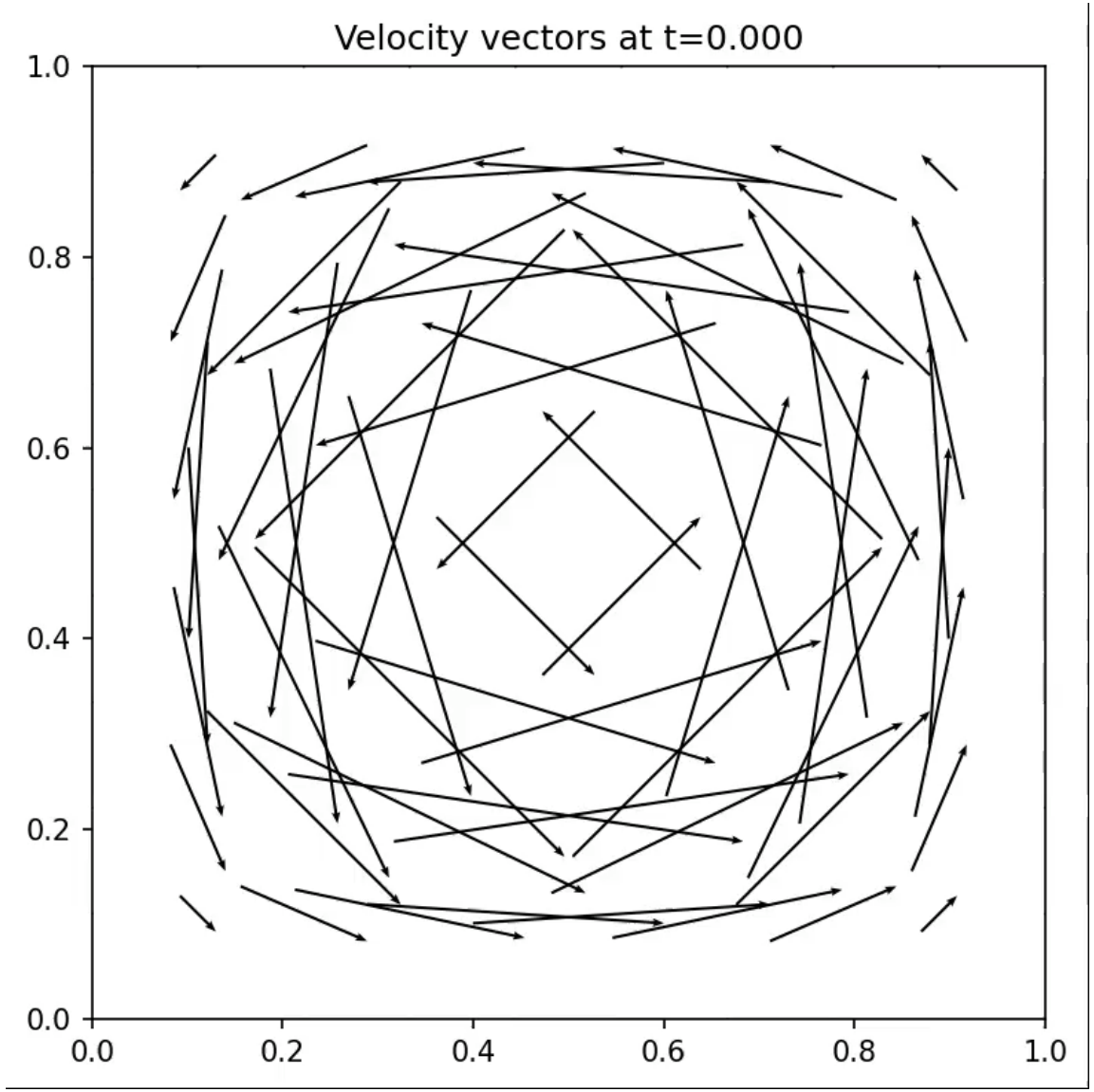


Another test case:

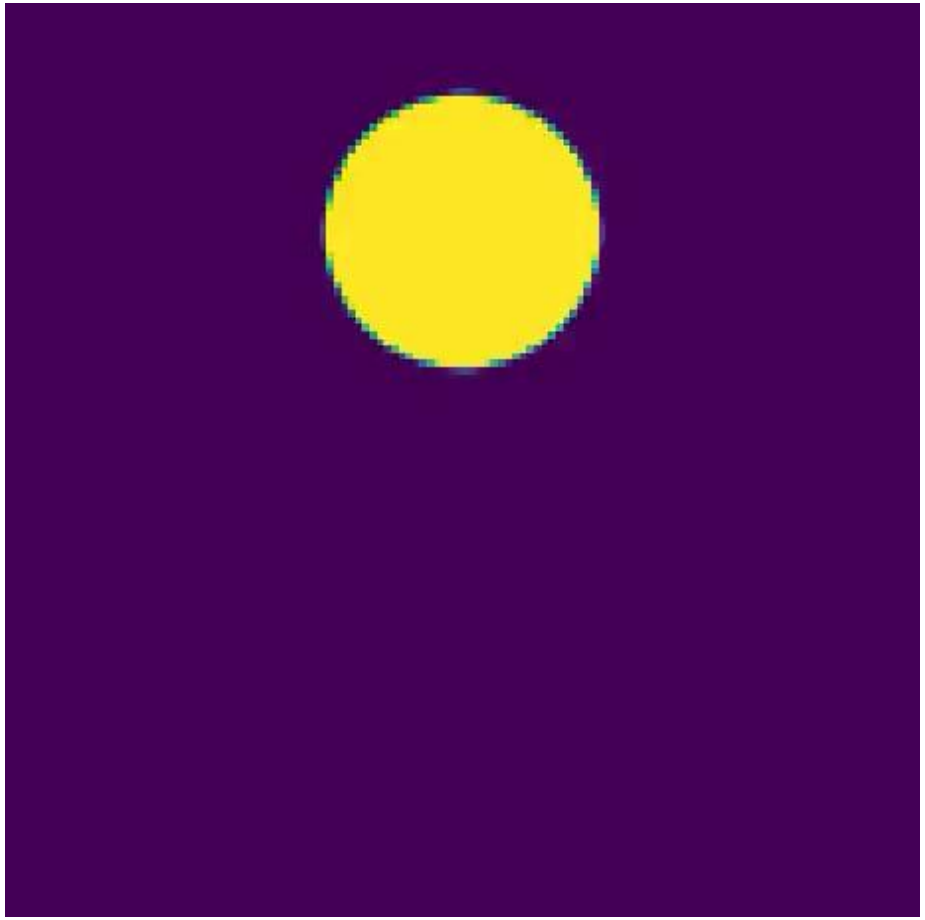
- Consider a oscillating, shearing flow field:

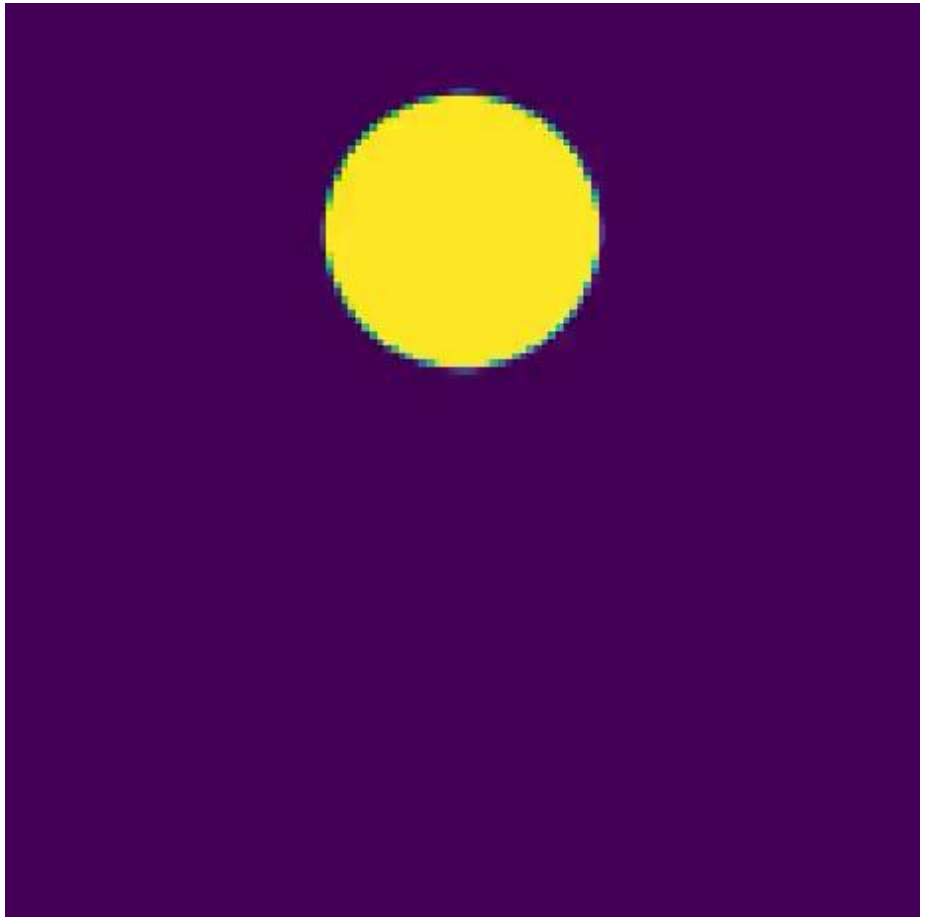
$$u = -\sin^2(\pi x)\sin(2\pi y)\cos\left(\frac{\pi t}{T}\right),$$

$$v = \sin(2\pi x)\sin^2(\pi y)\cos\left(\frac{\pi t}{T}\right)$$

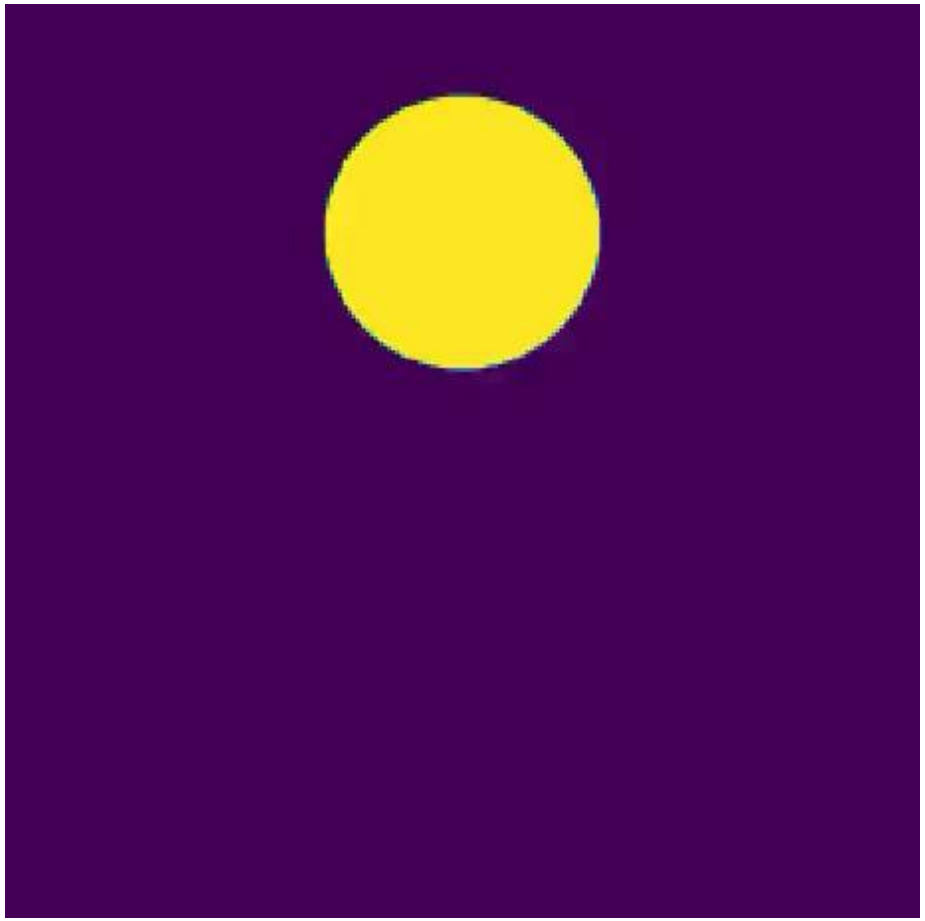


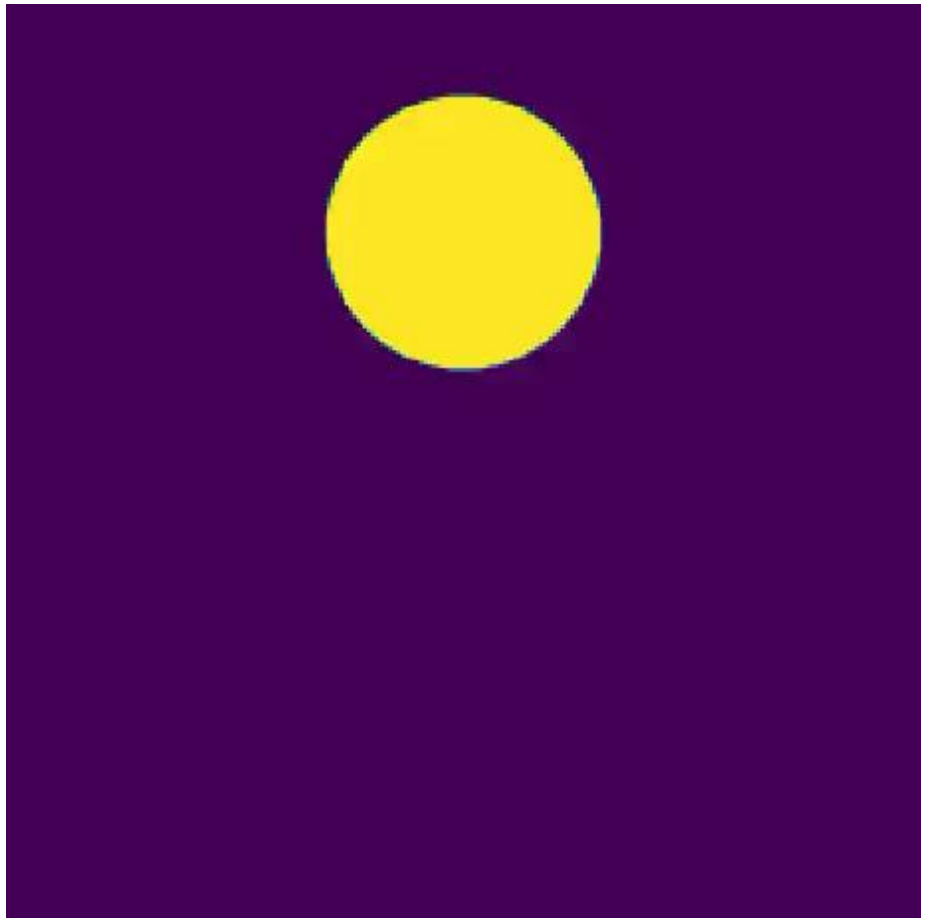
Another test case





Its more obvious on a finer mesh





Your Tasks:

- Write code to compute advection of void fraction as a passive scalar on a uniform, orthogonal mesh, using 1st order upwind scheme in space and 1st order explicit time stepping in time.
- Improve upon task 1 by adding CDI, see equation 1 in paper.
- Improve upon task 2 to use 2nd order spatial discretisation and 4th order temporal discretisation (RK4)
- Finally improve upon task 3 by implementing ACDI with 2nd order spatial discretisation: skew-symmetric-like splitting (equation 20 and 21)
- Compare the improvement of performance as each change is added. The two test cases in this paper are good choices, but you can come up with alternatives. Discuss the effect of changing Γ and ϵ

What to hand in

- Your code, in whatever language you prefer. Include both source code and compiled executable (if you are using c++ or Fortran etc.)
- Answers to the four tasks, you should include contour plots of ϕ and other relevant plots to show that your code is working correctly. Quantitative comparisons are useful as well, but not mandatory. All figures should be labelled clearly and discussion should be coherent, however there is no strict format requirement.